Amir Mehdi Soufi Enayati

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Summary

AI and Robotics researcher (Ph.D., Mechanical Engineering - Robotics) specializing in reinforcement learning, generative models, and intelligent control for autonomous systems. Developed RL pipelines, motion planners, and digital twins for robot manipulation and industrial automation. Led research projects, collaborated across disciplines, and mentored junior researchers, with publications in IEEE TRO, IEEE TAI, IEEE RA-L, and Neurocomputing. Interested in AI-assisted control, navigation, and perception of autonomous platforms.

Skills

AI & Machine Learning: Reinforcement Learning (Model-free, Multi-task, Safe, Sim2Real), Generative Models (Transformers, Diffusion Models, VAEs, GFlowNets), Representation Learning
Software & Tools: Python (PyTorch, Gymnasium, NumPy/SciPy), MATLAB/Simulink, Git, Bash, Docker
Robotics & Control: Simulation (PyBullet, Simscape, MuJoCo), Manipulation (Path Planning, Kinematics/Dynamics), Robust/Adaptive Control
Mechanical & Manufacturing: CAD (SolidWorks, CATIA), FEA (ABAQUS), DfM/DfA, Testing & Commissioning
Languages: English (Fluent), French (B1), Persian (Native)

Experience

Machine Learning Researcher ON, Canada (Jan. 2026 – Present)
Huawei Canada Research Institute

- Conducting Machine Learning research and developing high-fidelity physics-based simulations for robotic applications.

Postdoctoral Research Fellow ON, Canada (Oct. 2025 – Jan. 2026)
SickKids Research Institute (Molecular Medicine Program)

- Improved the sample efficiency of generative flow network models in drug discovery, and developed an uncertainty-aware multi-fidelity surrogate model.

Research Assistant | Co-instructor | Teaching Assistant BC, Canada (Sep. 2020 – Sep. 2025)
University of Victoria & UBC Okanagan (Advanced Control & Intelligent Systems Lab)

- Developed a transformer-based context inference module for zero-shot adaptation of RL agents to non-parametric task variations. (IEEE TNNLS Submission)
- Co-developed a generalizable guided generative motion planner to synthesize safe trajectories with new robots in new environments (RAS Submission).
- Developed a Sim2Real framework that exploits the intrinsic stochasticity of real-time simulation (IEEE TAI); and a symmetry-guided demonstration aggregation to accelerate RL training (IEEE TRO; NCAA).
- Co-developed a ready-to-deploy safe RL method for direct learning from physical robots (IROS 2024); and a high-fidelity industrial digital twin for closed-loop HIL testing (IEEE RA-L).
- Formulated a unified taxonomy of adaptive robotics using Markov Decision Processes (Neurocomputing)
- Co-instructed the graduate Applied ML course and TAed multiple undergrad courses.

Project Lead | Mechanical & Automation Engineer Tehran, Iran (Mar. 2016 – Mar. 2020)
Durali System Design & Automation (DSDA Co.)

- DSDA specializes in the design, development, and automation of custom control systems. Progressed from junior engineer to leading cross-functional teams from concept through commissioning.
- Led mechanical design for stage automation systems, including lifting platforms, rotating stages, and safety interlocks; supervised fabrication, installation, and commissioning.
- Served as team lead on a heavy-payload hydraulic 6-DOF motion simulator, including kinematic design, structural analysis, component selection, and integration of electro-hydraulic actuation with control hardware.
- Regularly interfaced with clients and suppliers, led technical reviews, and mentored junior engineers and technicians during installation and commissioning phases.

Mechanical Engineer, Intern Tehran, Iran (May 2015 – Dec. 2015)
S. Rezaei Research Center (SRRC)

- Member of an undergraduate design team building an electric motorcycle for the *5th Iranian Machine Design Competition*, contributing to mechanical design and prototyping; team won the **Best Design** award.

Education

Ph.D. Mechanical Engineering University of Victoria, BC, Canada (2020–2025)
- RL Sample-efficiency, Sim2Real, and Generalizability for Robot Manipulation

M.Sc. Mechanical Engineering – Control Systems Sharif University, Iran (2016–2018)
- Optimal Kinematics and Robust Control of Hydraulic Parallel Manipulator

B.Sc. Mechanical Engineering Sharif University, Iran (2012–2016)
- Design and Manufacturing of an Electromechanical Testbench

Selected Publications

- **Journal Papers**

- **A. M. S. Enayati**, H. Honari, K. Gupta, and H. Najjaran. Context Representation via Action-Free Transformer encoder-decoder for Meta Reinforcement Learning, Preprint available on **arXiv**.
- M. G. Tamizi, H. Honari, **A. M. S. Enayati**, A. Nozdryn-Plotnicki, and H. Najjaran. A Cross-Environment and Cross-Embodiment Path Planning Framework via a Conditional Diffusion Model, Preprint available on **arXiv**.
- **A. M. S. Enayati**, Z. Zhang, K. Gupta, and H. Najjaran. Sample-Efficient Reinforcement Learning with Symmetry-Guided Demonstrations for Robotic Manipulation, in **Springer Neural Computing & Applications**.
- Z. Zhang, J. Hong, **A. M. S. Enayati**, and H. Najjaran. Using Implicit Behavior Cloning and Dynamic Movement Primitive to Facilitate Reinforcement Learning for Robot Motion Planning, in **IEEE Transactions on Robotics**.
- **A. M. S. Enayati**, R. Dershan, Z. Zhang, D. Richert, and H. Najjaran. Facilitating Sim-to-Real by Intrinsic Stochasticity of Real-Time Simulation in Reinforcement Learning for Robot Manipulation, in **IEEE Transactions on AI**.
- **A. M. S. Enayati**, Z. Zhang, and H. Najjaran. A Methodical Interpretation of Adaptive Robotics: Study and Reformulation, in **Neurocomputing, 2022**.
- Z. Zhang, R. Dershan, **A. M. S. Enayati**, M. Yaghoubi, D. Richert, and H. Najjaran. A High-Fidelity Simulation Platform for Industrial Manufacturing by Incorporating Robotic Dynamics into an Industrial Simulation Tool, in **IEEE RA-L**.

- **Conference Papers**

- Y. Karpichev, T. Charter, J. Hong, **A. M. S. Enayati**, H. Honari, M. G. Tamizi, and H. Najjaran. Extended Reality for Enhanced Human-Robot Collaboration: a Human-in-the-Loop Approach, in **IEEE ROMAN 2024**.
- H. Honari, **A. M. S. Enayati**, M. G. Tamizi, and H. Najjaran. Meta SAC-Lag: Towards Deployable Safe Reinforcement Learning via MetaGradient-based Hyperparameter Tuning, in **IEEE IROS 2024**.
- N. Mahdian, M. Jani, **A. M. S. Enayati**, and H. Najjaran. Ego-Motion Aware Target Prediction Module for Robust Multi-Object Tracking, Preprint available on **arXiv**.
- J. Sol, **A. M. S. Enayati**, and H. Najjaran. Visual Deformation Detection Using Soft Material Simulation for Pre-training of Condition Assessment Models, in **IEEE CASE 2024**.

Patents

US Patent No. 2024/0003264 A1, Testing mechanical overspeed protection systems
M. Durali, M. H. Heydari, *A. M. S. Enayati*, S. M. Hosseini, 2024.

US Patent No. 2019/0292774 A1, Controlling acoustics of a performance space
M. Durali, M. A. Soleimani, F. F. Shabani, A. Habibollahi, A. Sohbatloo, *A. M. S. Enayati*, 2019.

Honors & Awards

Faculty of Graduate Studies Scholarship	University of Victoria (2021)
International Four-Year Doctoral Tuition Award	University of British Columbia (2020)
Graduate Deans Entrance Scholarship	University of British Columbia (2020)
Best Master Thesis Award for Technical Contributions	Sharif University (2018)
Best Electric Motorbike Design	5th Iranian Machine Design Contest (2015)
Member of Iran's National Elites Foundation	Iranian National Elites Foundation (2012)
Rank 10th in Physics & Mathematics	National University Entrance Exam (2012)
Silver Medalist	IPhO: Iran Physics Olympiad (2011)

References

Available upon request.